

Task 3 : Translation from Image

Introduction

Translation from Image project focuses on developing an intelligent system capable of extracting English text from images and videos and translating it into another language using a machine learning-based approach. With the growing need for automated document understanding, OCR and translation systems play a vital role in real-world applications such as digital archiving, surveillance analysis, assistive technology, and multilingual communication.

This system integrates Optical Character Recognition (OCR) with a neural machine translation model and a user-friendly graphical interface, ensuring both technical robustness and usability. The solution strictly processes English text, maintaining accuracy and preventing unintended multilingual translations.

Background

Optical Character Recognition (OCR) is a well-established computer vision technique that converts text present in images or videos into machine-readable format. Modern OCR systems leverage deep learning models to improve recognition accuracy under varying lighting, font styles, and orientations.

Machine Translation has evolved from rule-based systems to neural approaches such as Sequence-to-Sequence (Seq2Seq) models with attention mechanisms. Combining OCR with neural translation enables automatic extraction and translation of textual data from visual sources.

Learning Objectives

The key learning objectives of this project include understanding OCR pipelines, implementing neural machine translation models, and integrating machine learning models with graphical user interfaces. The project also aims to enhance practical knowledge of Python libraries such as EasyOCR, TensorFlow, OpenCV, and Tkinter.

Additional objectives include GPU optimization, preprocessing techniques, and deployment of ML models in real-time applications.

Activities and Tasks

The project was developed through structured activities including dataset preparation, OCR model integration, translation model training, GUI design, and system testing. Separate modules were created for OCR processing, translation inference, and interface interaction.

Tasks also included frame-wise video processing, English language filtering, and error handling for unsupported inputs.

Data :

Samples: 11071

	english	hindi
0	I have to go to sleep.	<start> मुझे सोना है। <end>
1	Muriel is 20 now.	<start> म्यूरियल अब बीस साल की हो गई है। <end>
2	Muriel is 20 now.	<start> म्यूरियल अब बीस साल की है। <end>
3	The password is "Muriel".	<start> कूटशब्द "Muriel" है। <end>
4	The password is "Muriel".	<start> पासवर्ड "Muriel" है। <end>
5	I will be back soon.	<start> मैं जल्द लौटूंगी। <end>
6	I'm at a loss for words.	<start> मैं तो लाजवाब हो गयी हूँ। <end>
7	This is never going to end.	<start> यह तो कभी खत्म न होगा। <end>
8	I just don't know what to say.	<start> मुझे नहीं पता मैं क्या कहूँ। <end>
9	That was an evil bunny.	<start> वह खरगोश दुष्ट था। <end>
10	Education in this world disappoints me.	<start> मैं इस दुनिया में शिक्षा पर बहुत निराश...
11	That won't happen.	<start> वैसा नहीं होगा। <end>
12	I miss you.	<start> मुझे तुम्हारी याद आ रही है। <end>
13	I miss you.	<start> मुझे आपकी याद आ रही है। <end>
14	You should sleep.	<start> तुम्हें सोना चाहिए। <end>
15	You should sleep.	<start> आपको सोना चाहिए। <end>
16	I never liked biology.	<start> मुझे जीव विज्ञान कभी भी पसंद नहीं था। ...
17	No I'm not; you are!	<start> मैं नहीं हूँ, तुम हो! <end>
18	That's MY line!	<start> वह तो मेरी लाईन है! <end>
19	Are you sure?	<start> पक्का? <end>

Model :

Layer (type)	Output Shape	Param #	Connected to
=====			
encoder_inputs (InputLayer)	[(None, 53)]	0	[]
decoder_inputs (InputLayer)	[(None, 59)]	0	[]
encoder_embedding (Embedding)	(None, 53, 300)	2226900	['encoder_inputs[0][0]']
decoder_embedding (Embedding)	(None, 59, 300)	2306700	['decoder_inputs[0][0]']
encoder_lstm (LSTM)	[(None, 53, 256), (None, 256), (None, 256)]	570368	['encoder_embedding[0][0]']
decoder_lstm (LSTM)	[(None, 59, 256), (None, 256), (None, 256)]	570368	['decoder_embedding[0][0]', 'encoder_lstm[0][1]', 'encoder_lstm[0][2]']
attention_layer (Attention)	(None, 59, 256)	1	['decoder_lstm[0][0]', 'encoder_lstm[0][0]']
..			
Total params: 9,618,794			
Trainable params: 9,618,794			
Non-trainable params: 0			

Testing :

```
Statement is : We didn't see anything.  
prediction is : हमने कुछ नहीं दिखा।  
  
Statement is : You have one chance.  
prediction is : आपके पास एक मौका है।  
  
Statement is : Spain is called "Espanya" in Catalan.  
prediction is : स्पेन को कैटलन में "एस्पान्या" कहते हैं।  
  
Statement is : I'm going to bed now.  
prediction is : मैं अभी तक जा रहा हूँ।  
  
Statement is : I haven't got in touch with him for a long time.  
prediction is : मैं उसे काफ़ी दिनों में सम्पर्क नहीं था।  
  
Statement is : Policemen work at the risk of their own lives.  
prediction is : पुलिस अफ़सर अपनी जानों को जोखिम में डालकर काम करते हैं।  
  
Statement is : She had plenty of acquaintances, but no friends.  
prediction is : उसके पास तो मैं बहुत दोस्त थे थे तो बहुत दोस्त बन रही थी।  
  
Statement is : Lava is dangerous.  
prediction is : लावा खतरनाक होता है।  
..  
prediction is : टीचर ने कहा, "बस के लिए इतना ही।"  
  
Statement is : Where is my dad?  
prediction is : मेरे पापा कहाँ हैं?
```

Skills and Competencies

This project strengthened skills in Python programming, machine learning model development, computer vision, and GUI-based application design. It enhanced problem-solving abilities related to text preprocessing, memory management, and modular software architecture.

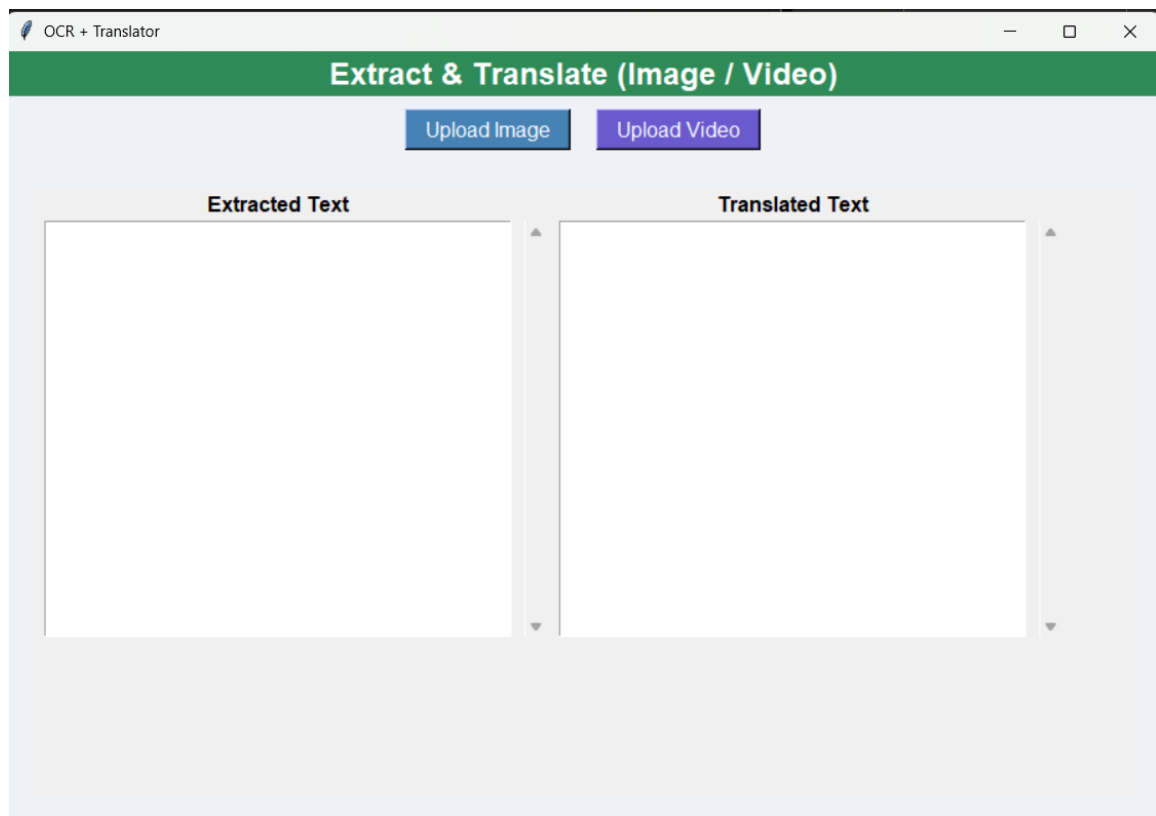
The project also improved experience with GPU acceleration and real-world ML deployment strategies.

Feedback and Evidence

Continuous testing and validation were conducted during development to ensure accurate OCR extraction and translation output. Feedback from test highlighted the clarity of the GUI and the efficiency of image and video processing.

Evidence of successful implementation includes extracted text samples, translated outputs, and GUI execution screenshots. The GUI output served as practical evidence of successful system integration. When invalid input was provided, the system correctly prompted the user to upload the text again, confirming that the constraint logic was functioning as intended.

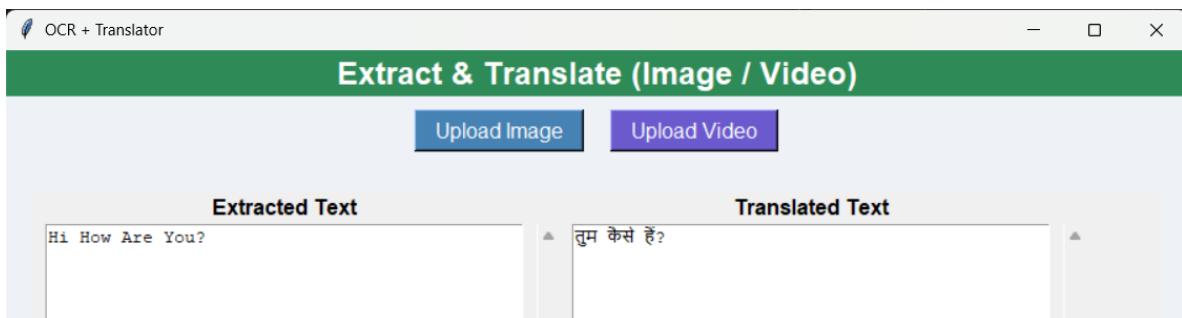
GUI :



Test image :



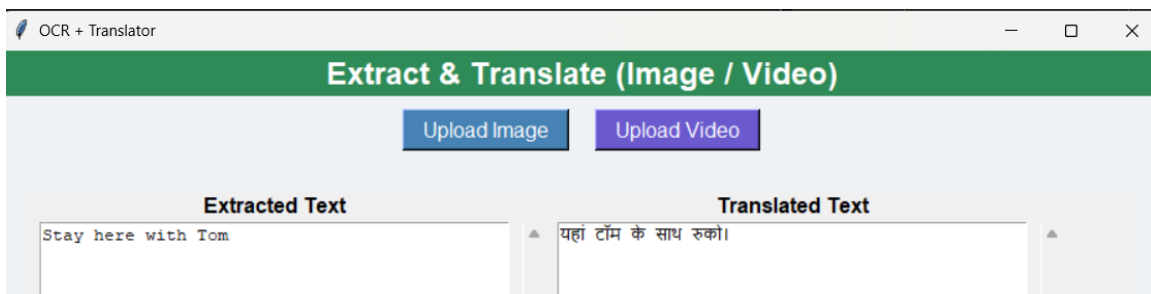
GUI Output :



Test Image :



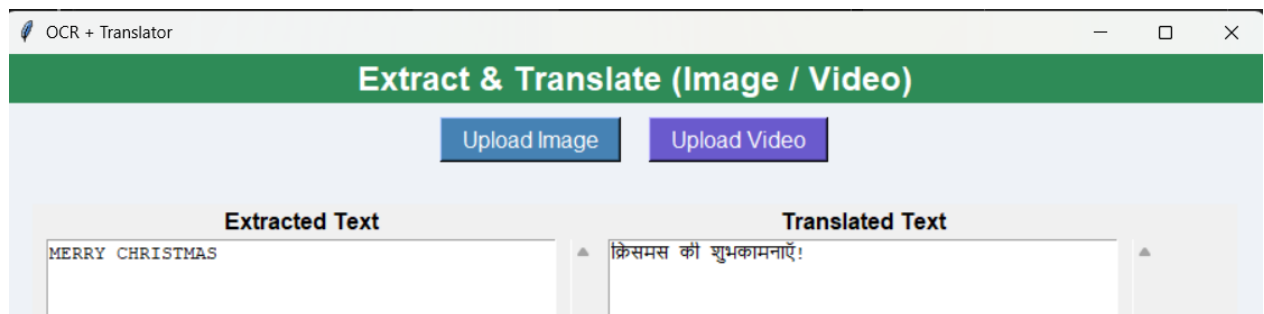
GUI Output :



Test Image :



GUI Output :



Challenges and Solutions

Major challenges included handling noisy image inputs, managing GPU memory, and avoiding duplicate text extraction in video frames. These issues were addressed through image preprocessing, memory growth configuration, and frame-step optimization.

Language filtering ensured that only English text was processed, improving translation accuracy.

Outcomes and Impact

The final system successfully extracts and translates English text from both images and videos with high accuracy. It demonstrates practical applicability in fields such as education, media analysis, and accessibility tools.

The project showcases the effective integration of OCR, machine learning, and GUI technologies into a single cohesive solution.

Conclusion

This project demonstrates a comprehensive approach to building an OCR-based translation system using machine learning. By combining EasyOCR, neural translation models, and a Tkinter-based GUI, the solution meets both functional and academic requirements.

Future enhancements may include multilingual support, real-time camera input, and cloud-based deployment.